

(System Software Fundamentals and Operating Systems)

2. (a) What do you mean by multiprogramming?
 (b) What is a scheduler? Explain its working using a diagram.
 (c) Consider a set of n tasks with known runtimes $r_1, r_2, \dots, r_n$ to be run on a uniprocessor machine. Which scheduling technique do you think will result in the maximum throughput, round robin, FCFS or SJF and why?
 (d) There are 3 CPU-intensive processes, which require 10, 20 and 30 time units and arrive at times 0, 2 and 6, respectively. How many context switches are needed if a shortest remaining time first scheduling algorithm is used? Do not count the context switches at time zero and at the end.

(e) Following jobs are to be executed on a single processor system and their order of arrival is as per the list :

Job - ID	CPU - Burst Time
p	4
q	1
r	8
s	1
t	2

Calculate the departure time (completion time) for job p if scheduling is round robin with time slice 1. 2+(2+2)+2+3+3

3. (a) What is binary semaphore? How producer-consumer relationship is implemented by using this semaphore?
 (b) Write the functions of 'fork()', 'wait()', and 'exit()' system calls.
 (c) Discuss the following loading scheme with their advantages and disadvantages :

- (i) Relocating loader
 (ii) Absolute loader. (1+4)+(2+1+1)+5

4. (a) What is Translation Lookaside Buffer (TLB)? Why is it needed?
 (b) Suppose the total size of main memory is 6MB, 1MB is occupied by the operating system and the rest part of the main memory is available for user space and it is divided into 5 equal fixed partitions. Find out the total internal fragmentation and external fragmentation for the following five jobs to be loaded into this main memory.

Jobs	Sizes
J1	450 KB
J2	1000 KB
J3	1024 KB
J4	1500 KB
J5	500 KB

- (c) What do you mean by 'Page fault'? What is Belady's Anomaly? 4+5+(2+3)

5. (a) Discuss briefly about SCAN disk scheduling and C-SCAN disk scheduling with suitable examples.

(b) Consider a system with five processes P_0 through P_4 and three resource types A, B, C. Suppose that, at a particular instance the system has been taken as follows :

Process	Allocation			MAX			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	7	5	3	3	3	2
P_1	2	0	0	3	2	2			
P_2	3	0	2	9	0	2			
P_3	2	1	1	2	2	2			
P_4	0	0	2	4	3	3			

Check whether the following permission is achieved or not by using Banker's algorithm. The process P_1 requests one additional instance of resource type A and two instances of resource type C.

- (c) Discuss, in brief, the 'semantic analysis' phase in a compiler. (3+3)+5+3